

NOTES AND REMARKS

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1. 13/5/09: Generation Process for V -Variable Fractals

In the following we see how one run of the backward process for generating V -variable fractals can be interpreted as *cell division to grow a single (V -variable) organism*.

On the other hand, we see how one run of the forward process for generating V -variable fractals can be interpreted as *interbreeding organisms to produce arbitrarily large families of (V -variable) organisms* whose empirical distributions approximate the natural distribution on V -variable fractals.

One run of either process is governed by an infinite sequence of V -tuples of transformation rules. For the backward process the transformations are interpreted as *cell replacement* rules. For the forward process the transformations are interpreted as *organism combination* rules. The results are related, but also very different.

1.1. Backward Process as Cell Division. See Figure 1 where $V = 4$.

Each cell is one of V possible types, where the integer V is fixed. Here we take $V = 4$ and types are indicated by the colours red, green, yellow and black. Organism growth is determined by an infinite sequence of V -tuples of transformations, here interpreted as *replacement* or *cell division* rules.

Begin with the initial cell at time $t = 0$. At times $t = 1, 2, 3, \dots$, cell division occurs according to a V -tuple of replacement rules as indicated in Figure 1. The kind of cell division, or equivalently the replacement rule applied to a cell, depends on the type of the cell, but the V -tuple of transformation rules depends on the external environment at the time of division.

If we *disregard* the information regarding individual cell type, we see the sequence of embryos at each division time converges to a limit adult organism (fractal). Cell type/colours are a book-keeping device and not part of the structure of the limit organism/fractal. See Figure 2 where the initial cell was of “green” type.

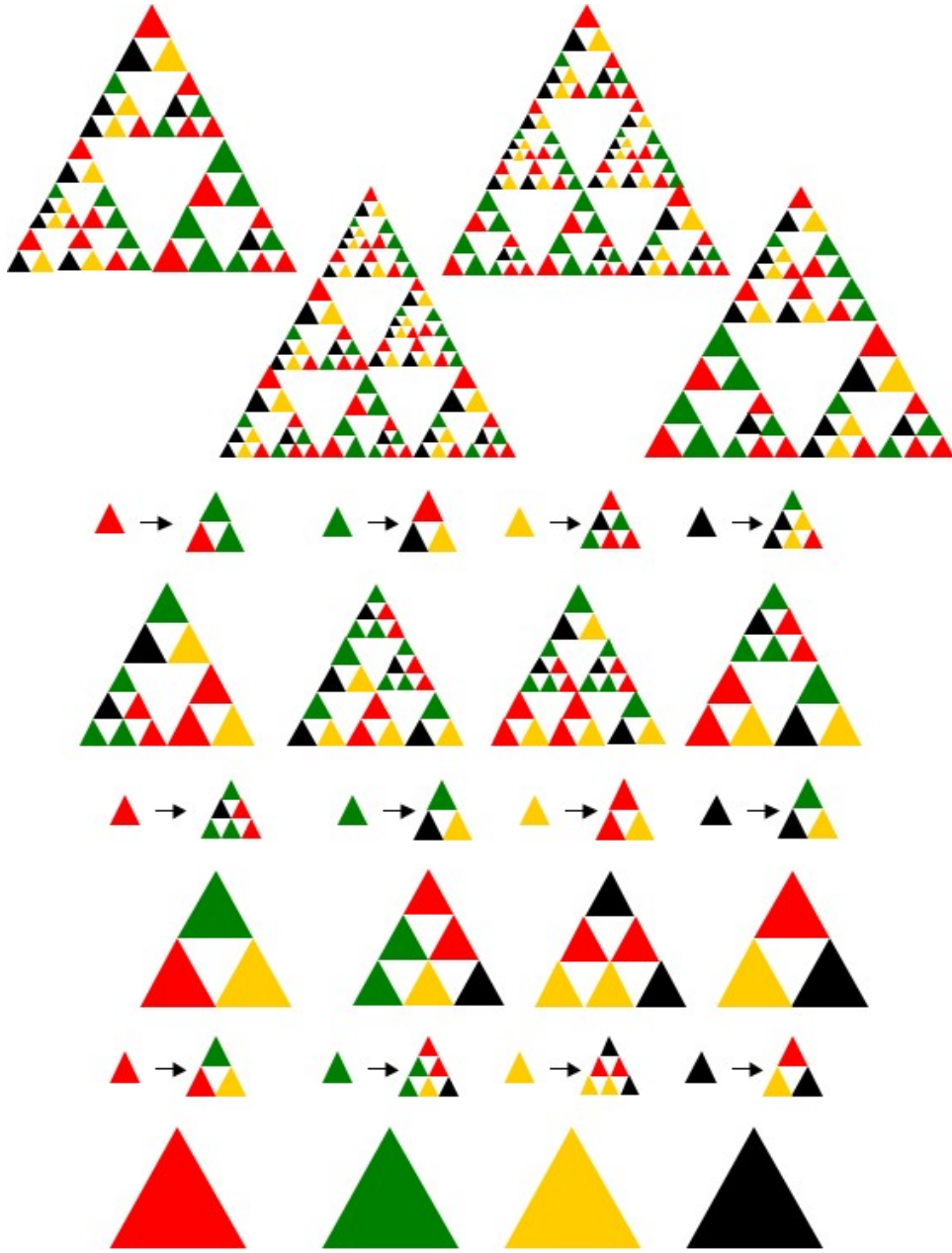


FIGURE 1. **Backward Process.** The transformations rules are interpreted as *re-
placement/division* rules.

The bottom line shows the four possible types for the initial cell. The division rules at time 1 are shown next and are followed by the 4 resulting organisms at time 1. The transformation rules at time 2 are then followed by the resulting organisms at time 2. Finally, the transformation rules at time 3 are followed by the resulting organisms at time 3.

Now observe any of the four growing organisms at successive times and ignore the cell type, i.e. colour. The organism structure at each time is a refinement of the organism at the previous time. This is more easily seen in Figure 2, where colours are suppressed and the development beginning from the green cell is shown.

1.2. Forward Process as Organisms Breeding. Here we work with entire organisms rather than individual cells. Types now refer to organisms and can again be regarded as a bookkeeping device. Each individual organism will be one of V possible types. See Figure 3 where $V = 4$ and the type of an individual organism is indicated by the colours red, green, yellow and black. An

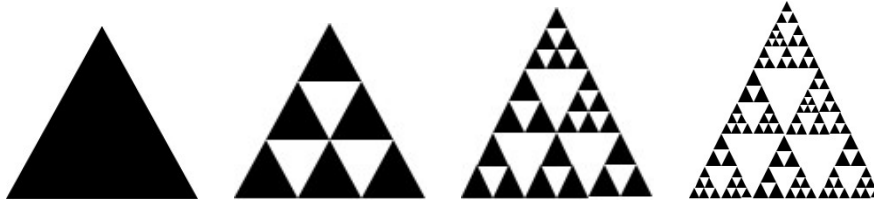


FIGURE 2. Organism development via cell division. Note how each embryo is a refinement of the embryo at the previous time.

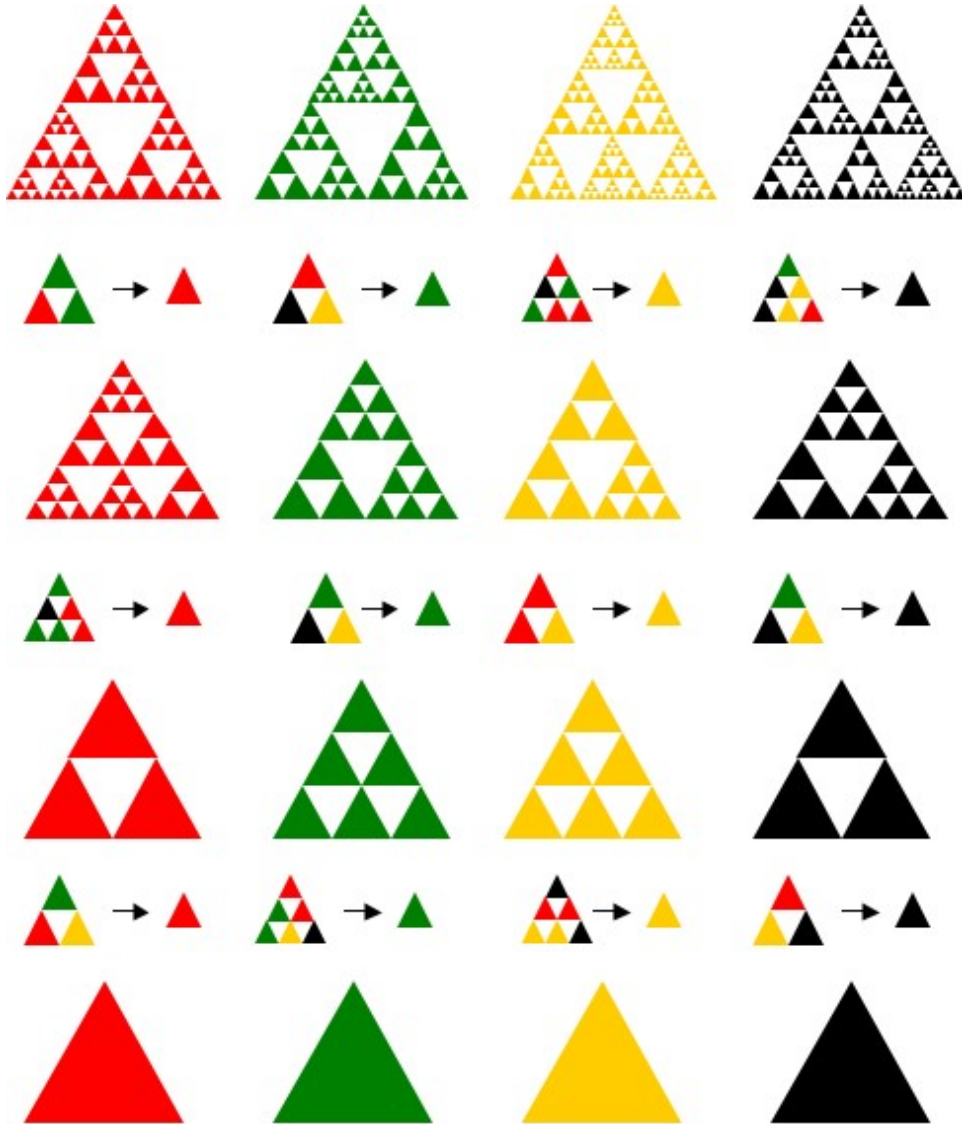


FIGURE 3. **Forward Process.** The transformations rules are interpreted as *combination* rules.

Begin with one individual organism of each type, ignoring any fine detail. The first set of combination rules produce 4 new organisms, one of each type. The second set produces another 4 and the third set produces another 4. After a few initial iterations, successive iterations produce a collection of organisms, 4 for each iteration, whose empirical distribution converges to the theoretical distribution of V -variable fractals.

entire *infinite family* of organisms is now generated by a single infinite sequence of V -tuples of transformations, where the transformations are now interpreted as *combination* or *reproduction* rules.

Begin with V individual organisms at time $t = 0$, one for each type. At times $t = 1, 2, 3, \dots$, breeding occurs according to a V -tuple of combination rules as indicated in Figure 1. At each generation there is one new organism produced for each type, as determined by the relevant transformation rule, and the V -tuple of transformation rules again depends on the external environment at the time of breeding.

We have used the same sequence of rules in Figure 3 as in Figure 1.

1.3. The Forward, Backward and Retarded Processes. *** Incomplete*** The *forward process* for $SG(2)$ consists of selecting a starting point x_0 and a sequence of maps $(g_i)_{i \geq 1}$ from $F = \{f_1, f_2, f_3\}$ in an iid manner according to some fixed probability distribution P . Then the sequence of points

$$x_0, x_1 = g_1(x_0), x_2 = g_2 \circ g_1(x_0), g_3 \circ g_2 \circ g_1(x_0), \dots,$$

converges a.s. to $SG(2)$. More precisely, the sequence of unit measures

$$\frac{1}{n}(\delta_{x_0} + \dots + \delta_{x_{n-1}})$$

converges in the weak sense of measures to the natural measure on $SG(2)$ induced from P .

The *backward process* says that for any initial non empty compact K , the sequence of sets

$$K, F(K), F^{\circ 2}(K), F^{\circ 3}(K), \dots \rightarrow SG(2),$$

in the Hausdorff sense, where

$$F(E) := f_1(E) \cup f_2(E) \cup f_3(E).$$

The *retarded* process takes sequences of the form:

$$x_0, x_1 = g_1(x_0), x_2 = g_1 \circ g_2(x_0), x_3 = g_1 \circ g_2 \circ g_3(x_0), \dots$$

Note the order of composition. Each sequence will converge to a point on $SG(2)$. Running the process many times gives a sample of points on $SG(2)$ whose empirical distribution converges a.s. to the natural distribution on $SG(2)$ given by P .

The previously described forward process for V -variable fractals is the forward process on the superfractal attractor. The backward process is really the retarded process on the superfractal attractor.

1.4. DNA. *** Incomplete*** A possible model for DNA reproduction is a variation of Section 1.2 as follows. ***The idea is to think of the organisms as a pair of DNA molecules. The allowable combinations correspond to crossing over. This needs to be formalised***

1.5. The Class of V -Variable Fractals. In the backward process we saw how a single organism (or fractal) was generated from an initial cell by a sequence of V -tuples of transformations chosen for example from a given collection $\mathfrak{F}$ of V -tuples of transformations. The collection of all possible V -variable fractals obtained in this manner is denoted by $\mathcal{K}_V$ and is called *the set of V -variable fractals*. Clearly,

$$\mathcal{K}_1 \subset \mathcal{K}_2 \subset \dots$$

The collection $\mathcal{K}_1$ is the collection of homogeneous fractals in the sense of Hambly. Moreover, $\mathcal{K}_V \uparrow \mathcal{K}_\infty$ as $V \rightarrow \infty$, where $\mathcal{F}_\infty$ is the collection of recursive fractals in the sense of Falconer, Graf, and Mauldin and Williams.

1.6. Contractive Maps and IFSs. Each of the 4 transformation rules in Figures 1 and 3 corresponds to an IFS for the Sierpinski gasket $SG(2)$ or $SG(3)$, if one disregards the colour information corresponding to the cell type or organism type. However, there is a convenient way to also incorporate this information.

Each 4-tuple of transformation rules corresponds to a map

$$(1) \quad \Phi : \left(\mathcal{K}(\mathbb{R}^2)\right)^4 \rightarrow \left(\mathcal{K}(\mathbb{R}^2)\right)^4, \quad \Phi(K_1, \dots, K_4) = \left(\Phi_1(K_1, \dots, K_4), \dots, \Phi_4(K_1, \dots, K_4)\right),$$

where $\mathcal{K}(\mathbb{R}^2)$ is the set of non empty compact subsets of $\mathbb{R}^2$. More precisely, identify the colours red, green, yellow and black respectively with the four coordinates in the natural order in (1). Let the IFSs for $SG(2)$ and $SG(3)$ be $F = (f_1, f_2, f_3)$ and $G = (g_1, \dots, g_6)$ respectively with notation as shown in Figure 4.

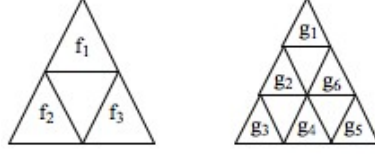


FIGURE 4. The IFS $F = (f_1, f_2, f_3)$ for $SG(2)$ and $G = (g_1, \dots, g_6)$ for $SG(3)$.

Then the map $\Phi^3 = (\Phi_1^3, \Phi_2^3, \Phi_3^3, \Phi_4^3)$, corresponding to the third 4-tuple of transformations in Figures 1 and 3, is given by

$$\begin{aligned} \Phi_1^3(K_1, K_2, K_3, K_4) &= f_1(K_2) \cup f_2(K_1) \cup f_3(K_2), \\ \Phi_2^3(K_1, K_2, K_3, K_4) &= f_1(K_1) \cup f_2(K_4) \cup f_3(K_3), \\ \Phi_3^3(K_1, K_2, K_3, K_4) &= g_1(K_1) \cup g_2(K_4) \cup g_3(K_2) \cup g_4(K_1) \cup g_5(K_1) \cup g_6(K_2), \\ \Phi_4^3(K_1, K_2, K_3, K_4) &= g_1(K_2) \cup g_2(K_4) \cup g_3(K_4) \cup g_4(K_3) \cup g_5(K_1) \cup g_6(K_3). \end{aligned}$$

The maps Φ^1 and Φ^2 corresponding to the first and second rows of transformations are defined similarly.

Let $\mathbf{A} = (A_1, A_2, A_3, A_4)$ denote the first 4-tuple of sets in Figures 1 or 3. Then for the backward and forward processes one obtains the sequence of sets

$$\begin{aligned} \mathbf{A}, \Phi^1(\mathbf{A}), \Phi^1 \circ \Phi^2(\mathbf{A}), \Phi^1 \circ \Phi^2 \circ \Phi^3(\mathbf{A}), \dots \quad & \text{(backward process),} \\ \mathbf{A}, \Phi^1(\mathbf{A}), \Phi^2 \circ \Phi^1(\mathbf{A}), \Phi^3 \circ \Phi^2 \circ \Phi^1(\mathbf{A}), \dots \quad & \text{(forward process).} \end{aligned}$$

The order of application of the maps is important. For the mapping $\Phi^1 \circ \Phi^2 \circ \Phi^3(\mathbf{A})$, the most “important” map is Φ^1 , since it determines features at the “top” level. The next most important is Φ^2 since it determines features at the second level of refinement, and so on. For the mapping $\Phi^3 \circ \Phi^2 \circ \Phi^1(\mathbf{A})$, the most important map is Φ^3 .

For the backward process each set is a refinement of the corresponding set at the previous time. For the forward process each set is a rescaled combination of copies of sets (usually more than one) from the previous time.

1.6.1. Using a family of standard IFSs. As in Figures 1 and 3, the most common way to generate the family $\mathfrak{F}$ of V -tuples of transformations is to begin with a family $\mathcal{F}$ of standard IFSs. Then $\mathfrak{F}$ is obtained by taking all possible actions of each IFS $F \in \mathcal{F}$ acting on all sets from arbitrary members of the previous V -tuple of sets.

For example, if $\mathcal{F} = \{(f_1, f_2, f_3), (g_1, \dots, g_6)\}$ then the possible component maps Φ_i for Φ are all those of the form

$$\Phi_i(K_1, \dots, K_V) = f_1(K_{i_1}) \cup f_2(K_{i_2}) \cup f_3(K_{i_3}) \quad \text{or} \quad g_1(K_{j_1}) \cup \dots \cup g_6(K_{j_6}),$$

where $i_1, i_2, i_3, j_1, \dots, j_6 \in \{1, \dots, V\}$.

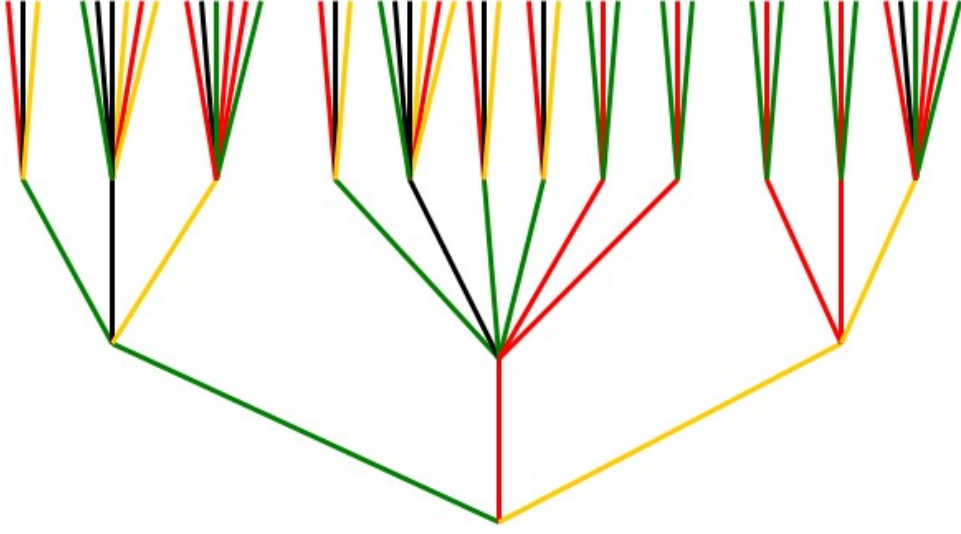


FIGURE 5. The construction tree corresponding to cell development as in Figure 1 for an initial red cell.

1.7. Construction Trees. Prefractal approximations to a V -variable fractal, or equivalently initial segments of the sequence of transformations, can be represented by trees. See Figure 5. The tree at each time is a subtree of the tree at each later time.

In the example corresponding to Figure 1 each node of the tree will correspond to an IFS $SG(2)$ or $SG(3)$, the edges coming out of the node will correspond to f_1, f_2, f_3 or $g_1, \dots, g_6$ respectively as in Figure 4 when reading the tree from left to right, and the branch colour have the same role as in Figure 1. In this case the labels at a node are determined by the number of edges coming out of the node, 3 for $SG(2)$ and 6 for $SG(3)$.

Note that at each level the subtree rooted at the upper end of each edge is the same for all edges of the same colour. This is essentially the V -variable feature of the fractals considered here.

We also note again that the colours are important in the construction process, but each prefractal is obtained from the corresponding tree without the colours playing a role.

1.8. Comparison with Other Notions. In order to compare with other notions in the literature we note the following.

- (1) *Homogeneous fractals:* These correspond to trees where $V = 1$, there is only one colour or type and so it no longer plays a role, and the subtrees rooted at each fixed level are the same. In particular, for each k there is either 3-fold branching or 6-fold branching at all nodes at level k .
- (2) *Recursive fractals:* These correspond to trees where there are no colours or types, and the subtree rooted at any node can be either 3-fold or 6-fold branching, without any constraint as a result of what happens at other nodes at the same or different levels.
- (3) *Recursive graph directed fractals:* Here we recall the construction in pages 8–12 of the book of Olsen.

One begins with a directed multigraph $(\mathcal{V}, E)$ where $\mathcal{V}$ is the set of vertices and E is the set of edges. Let E_u be the set of edges with initial vertex u . There are compact metric spaces X_u for each $u \in \mathcal{V}$. In Figure 1 the spaces X_u will correspond to 4 disjoint compact subsets of $\mathbb{R}^2$ containing the 4 prefractals constructed at each level of approximation. For each $u \in \mathcal{V}$ there is a collection of contractive maps

$$\mathcal{S}_u = \{S_e \mid e \in E_u, S_e : X_v \rightarrow X_u \text{ where } v \text{ is the terminal vertex of } e\}.$$

Using these ingredients one constructs the class of what we call the recursive graph directed fractals — the class of fractals which are realisations of random self-similar sets in the sense of Olsen. Each such fractal can be made to correspond to a (complicated) labelled construction tree.

The essential difference between recursive graph directed fractals and V -variable fractals is that in the latter case there are at most V distinct subtrees rooted at each level. The recursive graph directed fractals impose no such restriction, and in fact generalise the notion of a recursive fractal in (2) above.

It is the “ V distinct subtrees rooted at each level” property of V -variable fractals which allows one to run a forward process that approximates the class of all possible V -variable fractals. We discuss this more carefully later. There is a forward process for approximating a standard graph directed fractals, analogous to that for approximating a standard IFS fractal. But there is not a corresponding forward process which approximates the class of recursive graph directed fractals.

1.9. Probabilistic Version. *** Incomplete*** Now let P be a probability distribution on $\mathfrak{F}$. If $\mathfrak{F}$ is obtained from a family of IFSs as in Section 1.6.1 then

Suppose the V transformation rules (IFSs) at each generation are selected in an iid manner according to P , and independently of the rules selected at any other generation. This induces a probability distribution on the construction process and hence a probability distribution P_V on the set of generated fractals.

P_1 is the usual probability distribution on homogeneous random fractals. $P_V \rightarrow P_\infty$ in the weak sense as $V \rightarrow \infty$, where P_∞ is the standard distribution for random recursive fractals.

2. 11/2/10: Summary for V -variable Fractals

In the following fix a positive integer V and a family $\mathbf{F}$ of IFSs.

Definition 2.1. An *IFS tree* T is a tree with the following properties:

- (1) there is a single, level 0, root node $\emptyset$;
- (2) the *branching number* $n^{\mathbf{i}} \geq 2$ at each node $\mathbf{i}$ is finite and ≥ 2 ;
- (3) the edges with initial node $\mathbf{i}$ are numbered (“left to right”) from $1, \dots, n^{\mathbf{i}}$ and we write $\mathbf{i} = i_1 \dots i_k$ in the usual manner where $|\mathbf{i}| := k \geq 1$ is the *level* of $\mathbf{i}$;
- (4) there is an *IFS* $F^{\mathbf{i}} \in \mathbf{F}$ associated with each node $\mathbf{i}$, $n^{\mathbf{i}} = |F^{\mathbf{i}}|$ (the cardinality of $F^{\mathbf{i}}$), and the k th edge with initial node $\mathbf{i}$ is associated with the k th function in the IFS $F^{\mathbf{i}}$.

The unique compact set $K = K(T)$ associated with T in the usual manner is called a *V -variable fractal*.

Note that from the definition all trees are infinite.

For approximations to an IFS tree and the associated fractal, see Figure *** but *ignore* the “environments” and the symbols for the “types” $\{12, 3, 4\}$. In this example $\mathbf{F} = \{F(2), F(3)\}$ contains the IFSs generating $SG(2)$ and $SG(3)$ respectively, and the IFS at each node is determined by the branching number at the node. Looking at the top tree in the Figure, we see that at the base node the IFS is $F(3)$, and at the level one nodes, from left to right, the IFSs are $F(2), F(2), F(2), F(3), F(3), F(2)$.

The fractal is obtained by first applying $F(3)$ and it corresponds to $SG(3)$ at level one. At level 2, reading anticlockwise from the bottom left corner, the fractal is obtained by applying $F(2), F(2), F(2), F(3), F(3), F(2)$, and corresponds to $SG(2), SG(2), SG(2), SG(3), SG(3), SG(2)$.

Definition 2.2. Suppose V is a positive integer. A *V -variable IFS tree* is an IFS tree T with a *type* $\tau^{\mathbf{i}} \in \{1, \dots, V\}$ associated to each node $\mathbf{i}$, such that if two nodes at the *same* level k have the same type then:

- (1) both nodes have the same branching number and associated IFS, i.e.

$$|\mathbf{i}| = |\mathbf{j}| \ \& \ \tau^{\mathbf{i}} = \tau^{\mathbf{j}} \implies n^{\mathbf{i}} = n^{\mathbf{j}} \ \& \ F^{\mathbf{i}} = F^{\mathbf{j}};$$

- (2) comparable successor nodes have the same type as each other, i.e.

$$|\mathbf{i}| = |\mathbf{j}| \ \& \ \tau^{\mathbf{i}} = \tau^{\mathbf{j}} \ \& \ 1 \leq p \leq n^{\mathbf{i}} \implies \tau^{\mathbf{i}p} = \tau^{\mathbf{j}p}.$$

Remark 2.3.

- (1) A 1-variable IFS tree is essentially the same as an IFS tree which generate a scale irregular, i.e. homogeneous, fractal.
- (2) A V -variable IFS tree has at most V distinct IFSs associated to the nodes at each fixed level.

- (3) More strongly, if two nodes at the *same* level of a V -variable IFS tree have the same type then the subtrees rooted at these two nodes are identical, i.e.

$$|\mathbf{i}| = |\mathbf{j}| \ \& \ \tau^{\mathbf{i}} = \tau^{\mathbf{j}} \implies \sigma_{\mathbf{i}}T = \sigma_{\mathbf{j}}T.$$

In particular there are at most V distinct subtrees rooted at the *same* level. $\square$

The following is used in the construction and analysis of V -variable fractals.

Definition 2.4. An *environment* $\mathcal{E}$ assigns to each type $v \in \{1, \dots, V\}$ an IFS $F = F_v \in \mathbf{F}$ and a sequence of types $(\tau_{v1}, \dots, \tau_{vn})$, where $n = |F_v|$. We write

$$\mathcal{E} = (E_1^\mathcal{E}, \dots, E_V^\mathcal{E}), \quad E_v^\mathcal{E} = (F_v^\mathcal{E}, \tau_{v1}^\mathcal{E}, \dots, \tau_{vn}^\mathcal{E}).$$

Remark 2.5 (Construction of V -variable IFS trees). V -variable IFS trees are constructed recursively as follows:

- (1) *Stage 0:* Assign a type $\tau^\emptyset$ to the root node $\emptyset$.
- (2) Assume that by the completion of stages 0 to $n-1$:
 - (a) the tree and all nodes $\mathbf{i}$ with $|\mathbf{i}| \leq n-1$ have been constructed,
 - (b) a type $\tau^{\mathbf{i}}$ has been assigned if $|\mathbf{i}| \leq n-1$,
 - (c) an IFS $F^{\mathbf{i}}$ has been assigned if $|\mathbf{i}| \leq n-2$,
 - (d) the two properties in Definition 2.2 hold if $|\mathbf{i}| \leq n-2$.
- (3) *Stage n :* Choose an environment $\mathcal{E}$. We follow the notation of Definition 2.4 and use $\mathcal{E}$ in the obvious way to assign a type to the level $n-1$ nodes, to construct the level n nodes, and to assign a type to every level n node.

More precisely, for each node $\mathbf{i}$ with $|\mathbf{i}| = n-1$:

- (a) if $v = \tau^{\mathbf{i}}$ is the type already assigned to $\mathbf{i}$, then $F^{\mathbf{i}} := F_v^\mathcal{E}$ is the IFS assigned to $\mathbf{i}$,
- (b) the branching number at $\mathbf{i}$ is $|F^{\mathbf{i}}|$,
- (c) for $1 \leq u \leq |F^{\mathbf{i}}|$ the type $\tau^{i\mathbf{u}} := \tau_{vu}^\mathcal{E}$ is the type assigned to the node $i\mathbf{u}$.

It follows by an easy induction that the properties in Definition 2.2 hold at all nodes. $\square$

Definition 2.6. The environment $\mathcal{E}$ in Definition 2.4 is a *neck* if all $\tau_{uv}^\mathcal{E}$ are equal.

A *neck* occurs at level n in the construction of a V -variable IFS tree if the environment $\mathcal{E}$ applied at stage n is a neck environment.

If a neck occurs at level n then the type assigned to every node at that level is the same. It follows that the IFS assigned to every node at level n will be the same, and moreover all (infinite) subtrees rooted at level n will be the same. But the subtrees themselves are only constructed at later stages, and even the common value of the IFS at each level n node is not determined until stage $n+1$.

There is no restriction on the IFSs occurring in the environment $\mathcal{E}$; these IFSs are applied at level $n-1$.

Definition 2.7. Fix a probability distribution P on $\mathbf{F}$. This induces a probability distribution P_V on the set of environments $\mathcal{E}$ by first choosing the IFSs $F_v^\mathcal{E}$ for $v \in \{1, \dots, V\}$ in an iid manner according to P . The types $\tau_{uv}^\mathcal{E}$ for the same range of v and for $1 \leq u \leq |F_v^\mathcal{E}|$ are chosen in an iid manner according to the uniform distribution on $\{1, \dots, V\}$ and independently of the $F_v^\mathcal{E}$, except that the cardinality of the set of such types depends on the cardinality of the chosen IFSs.

The probability distribution on the set of V -variable IFS trees is obtained by choosing $\tau^\emptyset$ according to the uniform distribution and independently choosing the environments at each stage in an iid manner according to P_V . This probability distribution on IFS trees induces a probability distribution of V -variable fractals. Both the distributions on trees and on fractals are also denoted by P_V .

Note that although the distribution P_V on environments is a product measure, this is very far from the case for the corresponding distribution on trees and on fractals. There is a very high degree of dependency between the types (and hence the IFSs) assigned to two different nodes at the same level.

Note also that necks will occur infinitely often

3. 8/4/09: Analysis on Fractals via Flat Chains

Could one use the flat chain setting (end of my 81 paper), together with boundary operators as in Peres and Lyons book, "PROBABILITY ON TREES AND NETWORKS", section 2.4, to develop analysis on pcf and other fractals?

4. 7/11/09: Evolving Galaxies

Could one use the ideas of optimal transport in Ch 8 of Villani, Topics in Optimal Transportation, to get a fractal limiting distribution?

REFERENCES

- [BHS09] M. F. Barnsley, John E. Hutchinson, and Örjan Stenflo, *V-variable Fractals: Dimension results* (2009). submitted for publication.
- [Sce07] Bob Scealy, *V-variable fractals and interpolation*, 2007. draft Version, 7 September.
- [Sce08] ———, *V-variable fractals and interpolation*, 2008.